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THE SCHOOL OF NATURAL SCIENCES
THE DEPARTMENT OF MATHEMATICS,
STATISTICS AND ACTUARIAL SCIENCE**

**TITLE: A Quantitative Assessment of Market Efficiency on
LUSE: Evidence from Econometric and Statistical Testing**

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Finally, I am grateful to my family and friends for their unwavering support, patience, and motivation throughout my studies.

Dedication

I dedicate this research proposal to my family, whose unwavering love, patience, and encouragement have been the foundation of my journey. To my parents, for their sacrifices and for instilling in me the values of hard work, integrity, and perseverance. To my siblings, for being my constant source of motivation and laughter through challenging times.

I also dedicate this work to my lecturers and mentors, whose guidance and dedication to knowledge have shaped my academic and personal growth. Their insights and encouragement have inspired me to approach every challenge with curiosity and resilience.

Lastly, I dedicate this proposal to all my friends and colleagues who have walked alongside me throughout this journey, for their support, shared struggles, and belief in my potential. This work stands as a reflection of the collective strength, inspiration, and faith of all those who have contributed to my path.

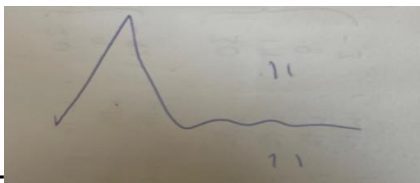
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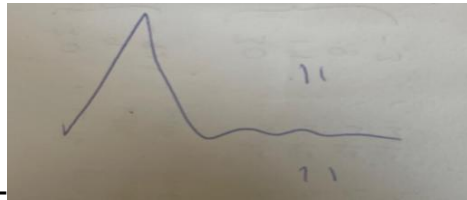


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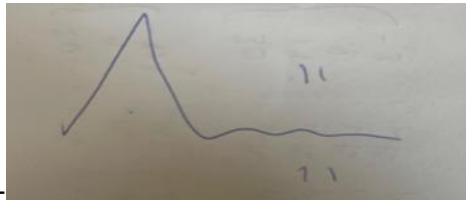
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Date 29th October, 2025

Abstract

The efficiency of capital markets is fundamental to fostering investor confidence, encouraging participation, and ensuring the fair valuation of securities. In Zambia, the Lusaka Securities Exchange (LUSE) has experienced substantial growth in both size and trading activity, reflecting increased investor interest and improved market awareness. However, there is limited recent empirical evidence assessing whether this progress has translated into greater market efficiency. This study aims to assess the weak-form efficiency of LUSE by examining whether stock prices follow a random walk, as postulated by the Efficient Market Hypothesis (EMH). Using a quantitative research approach, the study analyzes daily and weekly stock price data from selected companies listed on LUSE over the period January 2022 to October 2025, representing the most recent and complete dataset currently available. Statistical techniques, including the Runs Test, Autocorrelation Test, and Variance Ratio Test, are employed to evaluate the randomness and predictability of stock returns. These tests jointly determine whether successive price changes are independent, thereby indicating the degree to which past information influences current price movements. Data analysis is conducted using R and Stata software at a 5% significance level to ensure robustness and accuracy. The findings of this study will establish whether investors can exploit historical price information to earn abnormal returns or whether stock prices fully and instantaneously reflect all publicly available information. By providing updated empirical evidence based on recent LUSE performance, this research contributes to the understanding of Zambia's capital market dynamics and offers valuable insights for investors, policymakers, and regulators seeking to enhance transparency, liquidity, and alignment with international market standards.

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List Of Acronyms

Acronym / Symbol	Meaning
LUSE	Lusaka Securities Exchange
EMH	Efficient Market Hypothesis
ZSE	Zimbabwe Stock Exchange
ACF	Autocorrelation Function
VR	Variance Ratio
H ₀	Null Hypothesis
H ₁	Alternative Hypothesis
R	Statistical software (R Programming Language)
μ	Expected mean of runs (in Runs Test formula)

CHAPTER ONE: INTRODUCTION TO THE STUDY

1.0 Background of the Study

The efficiency of capital markets is fundamental to the functioning of modern financial systems, underpinning investor confidence, effective price discovery, and resource allocation. An efficient market ensures that asset prices fully and accurately reflect all available information, minimizing opportunities for arbitrage and enabling investors to make informed decisions. Conversely, inefficiencies can lead to mispricing, market volatility, and speculative trading, potentially undermining economic growth and investor trust. Understanding the efficiency of a financial market is therefore critical for investors, regulators, and policymakers.

The Efficient Market Hypothesis (EMH), formalized by Fama (1970), categorizes market efficiency into three forms: weak, semi-strong, and strong. Weak-form efficiency asserts that stock prices reflect all historical price and volume information, implying that past data cannot be used to predict future returns consistently. Semi-strong efficiency suggests that prices quickly incorporate all publicly available information, while strong-form efficiency extends this to include private or insider information. Among these, weak-form efficiency provides the foundation for analyzing short-term price movements and the applicability of technical trading strategies.

Emerging markets, including many African stock exchanges, often face structural challenges that can limit efficiency. These include low liquidity, limited investor participation, narrow product offerings, thin trading volumes, and imperfect regulatory frameworks. Such conditions can distort price signals, increase return volatility, and reduce the ability of markets to accurately reflect underlying economic conditions. Despite these limitations, African stock exchanges have experienced steady growth in activity, reflecting increasing investor interest, broader financial development, and a gradual improvement in market infrastructure.

The Lusaka Securities Exchange (LUSE), Zambia's main equity market, exemplifies both the potential and constraints of frontier markets. In recent years, LUSE has witnessed significant growth in market activity. The LUSE All Share Index increased from approximately 17,000 points to 25,000 points within ten months, while trading volumes rose from under 5,000 trades in 2020 to over 31,800 trades in 2024. This growth signals heightened investor confidence, expanding market participation, and an evolving capital market environment. The increase in both domestic and foreign investment activity highlights the strategic importance of LUSE in Zambia's broader financial system.

Despite this progress, there remains a lack of current, rigorous research on the efficiency of LUSE. Most existing studies on African stock markets have concentrated on larger or more liquid exchanges such as South Africa, Kenya, and Zimbabwe, leaving smaller frontier markets like Zambia relatively underexplored. Where LUSE has been studied, research is either outdated or limited in scope, often employing short datasets or non-robust statistical methodologies. Consequently, there is uncertainty regarding whether LUSE stock returns follow a random walk,

indicating weak-form efficiency, or whether predictable patterns exist that could be exploited by investors.

Market efficiency has practical implications. For investors and portfolio managers, knowledge of market efficiency informs the choice of investment strategies, including the effectiveness of technical versus fundamental analysis, risk management, and diversification approaches. For policymakers and regulators, understanding efficiency supports efforts to enhance transparency, improve market infrastructure, and protect investors. Efficient markets also attract foreign investment, stimulate economic growth, and promote financial stability. Comparative studies in African markets further underscore the relevance of examining LUSE. For instance, Mlambo and Biekpe (2007) found mixed evidence of weak-form efficiency across African exchanges, while Chowa, Nyanhete, and Mhlanga (2014) demonstrated that structural reforms significantly impacted market efficiency in Zimbabwe. Such studies indicate that market structure, liquidity, and institutional quality are crucial determinants of efficiency, making LUSE a valuable case for empirical investigation.

This study aims to fill this research gap by quantitatively assessing the weak-form efficiency of LUSE. Using daily and weekly stock price data from selected companies over the period January 2022 – October 2025, statistical and econometric tests—including the Runs Test, Autocorrelation Test, and Variance Ratio Test—will be employed to examine whether stock returns follow a random walk. The results are expected to provide evidence on LUSE’s efficiency, guiding investors, asset managers, and policymakers in their decision-making, and contributing to academic knowledge on frontier market behavior.

1.1 Statement of the Problem

The ideal financial market adheres to all three forms of EMH, ensuring rapid dissemination of information, absence of insider trading, and fully reflective asset prices. In such markets, consistent abnormal returns cannot be achieved without assuming additional risk. The LUSE, however, presents several characteristics that may impede this ideal. It lists only 20 companies, many of which are thinly traded, limiting the market’s ability to represent broader economic dynamics. Delayed reporting and limited disclosure practices reduce transparency and contribute to information asymmetry.

Low liquidity and constrained trading activity are particularly problematic in emerging markets like LUSE, leading to nonsynchronous trading where securities do not trade simultaneously. This can create serial correlation in returns, violating weak-form efficiency. Limited market breadth, concentrated ownership, and narrow product offerings further reduce effective price discovery, making it challenging to assess the true value of listed companies. Trading hours, limited to weekdays between 9:00 and 14:00, compound these constraints, restricting participation and reducing market responsiveness to information.

The regulatory environment, though improving, still faces challenges in enforcement and timely dissemination of information. These conditions may encourage investors and asset managers to

rely on technical analysis or trading strategies without knowing if consistent abnormal returns are achievable. Consequently, investment decisions may be suboptimal, market pricing may not reflect intrinsic values, and investor confidence could be undermined. Policymakers, too, lack up-to-date empirical evidence to guide reforms, hampering efforts to develop a transparent and robust capital market.

Given these challenges, it is essential to examine whether LUSE exhibits weak-form efficiency. Establishing this will provide insights into market behavior, inform investment strategies, and support policy decisions aimed at enhancing market functionality, transparency, and investor protection.

1.2 Significance of the Study

This study has both theoretical and practical significance. Theoretically, it contributes to the literature on market efficiency in frontier markets, particularly in the African context, where empirical evidence remains limited. By focusing on LUSE, the research provides a current, data-driven evaluation of weak-form efficiency, complementing studies conducted in Zimbabwe, Botswana, South Africa, and other African markets.

Practically, the study informs investors, asset managers, and policymakers. For investors, understanding market efficiency aids in selecting trading strategies, evaluating risk, and diversifying portfolios. Asset managers can align investment decisions with evidence on price predictability, while regulators and policymakers can identify areas for reform, such as improving market transparency, expanding product offerings, and enhancing enforcement of disclosure requirements. Overall, the study supports sustainable capital market development and fosters investor confidence in Zambia.

1.3 Research Aim

The aim of this study is to evaluate the weak-form efficiency of the Lusaka Securities Exchange (LUSE) by examining whether stock price movements follow a random walk pattern. Through statistical and econometric tests, the study seeks to determine the extent to which past price information influences current stock returns, thereby assessing the predictability and informational efficiency of the Zambian capital market.

1.4 Specific Objectives

1. To assess whether stock returns on the Lusaka Securities Exchange exhibit weak-form efficiency using statistical and econometric tests.
2. To provide recommendations for investors, asset managers, and policymakers based on the findings of the study.

1.5 Research Questions

1. Do stock returns on the Lusaka Securities Exchange follow a random walk, indicating weak-form market efficiency?
2. What are the implications of the efficiency or inefficiency of LUSE for investors, asset managers, and policymakers?

1.6 Justification of the Study

Given the limited research on Zambia's capital market and its growing economic significance, this study is justified on both practical and academic grounds. Practically, findings will help investors make informed decisions, reduce risks associated with mispricing, and optimize portfolio management strategies. Policymakers can leverage the results to improve market regulations, transparency, and investor protection. Academically, the study addresses a critical research gap, extending the literature on African frontier market efficiency by focusing on Zambia's context and employing rigorous statistical methods. Overall, this study is timely as it provides updated empirical evidence on LUSE's efficiency, guiding both academic inquiry and policy formulation aimed at strengthening Zambia's financial markets

1.7 Scope of the Study

This study focuses exclusively on the weak-form efficiency of the Lusaka Securities Exchange. It examines daily and weekly stock price data for companies listed on LUSE from January 2022 and October 2025. The research applies quantitative methods, including runs tests, autocorrelation tests, and variance ratio tests, to evaluate whether stock returns follow a random walk. The study does not examine semi-strong or strong forms of market efficiency, nor does it analyze macroeconomic variables' direct impact on returns, though such factors may be noted as contextual influences. The findings will therefore be limited to the period and data scope defined, and generalization beyond LUSE should be made cautiously.

1.8 Limitations of the Study

While this study aims to provide a robust assessment of the weak-form efficiency of the Lusaka Securities Exchange (LUSE), several limitations are acknowledged that may affect the findings and their generalizability.

Firstly, the dataset spans only from January 2022 to October 2025, which restricts the ability to observe long-term trends in efficiency and to compare pre- and post-reform phases of the market. The shorter time horizon may also amplify temporary shocks or cyclical patterns in the data, such as those associated with post-pandemic recovery, inflationary pressures, or changes in investor sentiment.

Secondly, the relatively small number of listed and actively traded companies on LUSE constrains the breadth of the analysis. Thin trading and nonsynchronous price movements can induce autocorrelation and distort statistical test outcomes.

Lastly, the study's focus on weak-form efficiency excludes examination of semi-strong and strong-form efficiency. As a result, the findings will provide insights only on whether historical price and volume data can predict returns, without addressing how quickly markets incorporate public or private information. Additionally, external macroeconomic shocks, policy changes, or global market events during the study period may affect stock prices independently of intrinsic market efficiency, introducing variability that is difficult to quantify.

Despite these limitations, the combination of multiple statistical tests enhances the reliability of findings, ensuring a balanced assessment of weak-form efficiency on LUSE.

1.9 Definition of Key Terms

- **Weak-Form Market Efficiency:** A form of the Efficient Market Hypothesis (EMH) which asserts that all past price and volume information is already reflected in current stock prices, implying that historical data cannot be used to predict future returns.
- **Random Walk:** A statistical process suggesting that stock price changes are independent of each other and unpredictable, consistent with weak-form efficiency.
- **Lusaka Securities Exchange (LUSE):** Zambia's principal stock exchange, established in 1993, responsible for facilitating equity trading, promoting investment, and supporting capital market growth in the country.

1.10 Organization of the Study

The study is structured into five chapters, each addressing a specific aspect of the research process and analysis.

1. **Chapter One – Introduction:** Provides the background, research problem, objectives, significance, scope, limitations, and organization of the study.
2. **Chapter Two – Literature Review:** Synthesizes theoretical frameworks, prior empirical studies, and key determinants of market efficiency in African and global contexts.
3. **Chapter Three – Research Methodology:** Details research design, data sources, sampling strategy, and statistical and econometric techniques employed.
4. **Chapter Four – Results and Discussion:** Presents findings from descriptive and inferential statistical analyses, interpreting results in the context of weak-form efficiency.
5. **Chapter Five – Conclusion and Recommendations:** Summarizes key findings, draws conclusions on LUSE's weak-form efficiency, and provides actionable recommendations for investors, policymakers, and regulators.

CHAPTER TWO: LITERATURE REVIEW

2.0 Introduction

The literature review provides both the theoretical and empirical foundation for examining the weak-form efficiency of the Lusaka Securities Exchange (LUSE). It situates the research within the broader context of capital market studies and highlights the need for updated empirical evidence specific to Zambia. By reviewing the Efficient Market Hypothesis (EMH) and analyzing prior studies from developed, emerging, and frontier markets, this chapter identifies structural, institutional, and methodological factors that influence market efficiency. These insights underscore the relevance of the current study, which seeks to evaluate whether LUSE stock prices follow a random walk, thereby testing the weak-form EMH.

The weak form of EMH, central to this study, posits that stock prices fully incorporate all past trading information, including historical prices and volumes. Consequently, no investor can consistently achieve abnormal returns using technical analysis alone (Fama, 1970; Fama, 1991). In frontier markets like Zambia, however, structural constraints such as limited listings, low liquidity, and weak regulatory oversight may lead to deviations from ideal market efficiency. This review critically examines theoretical frameworks and empirical findings while identifying methodological approaches that inform the current research design.

2.1 Theoretical Framework: Efficient Market Hypothesis

The Efficient Market Hypothesis (EMH) remains the cornerstone of modern financial theory, providing the theoretical foundation for understanding price behavior in capital markets. Eugene Fama's seminal work in the 1960s and 1970s laid the groundwork for this concept, formalizing the idea that financial markets are "informationally efficient" (Fama, 1970; Fama, 1991). EMH posits that security prices fully and instantaneously reflect all available information, implying that investors cannot consistently earn abnormal returns by exploiting publicly available or historical information. This principle directly informs the study of weak-form market efficiency, which examines whether past price and volume data can predict future stock returns.

Fama's (1970) pioneering review categorized market efficiency into three distinct forms: **weak, semi-strong, and strong**. The weak form asserts that all historical trading information—including past prices, returns, and trading volumes—is fully incorporated into current prices, making technical analysis ineffective. The semi-strong form extends this notion to all publicly available information, such as earnings announcements, dividend declarations, and macroeconomic indicators. Finally, the strong form argues that even insider or private information is reflected in stock prices, a condition rarely observed in practice due to legal and ethical constraints (Fama, 1991).

The theoretical rationale for EMH rests on the interplay of three key market mechanisms: (i) the rapid dissemination of information to all market participants, (ii) the active engagement of rational, profit-maximizing investors who exploit any temporary mispricing, and (iii) the continuous adjustment of prices as investors incorporate new information (Fama, 1970; Lo & MacKinlay, 1988). In an efficient market, these mechanisms ensure that prices at any given time represent the collective knowledge and expectations of market participants. For the weak form, this means that stock returns follow a **random walk**, characterized by the property that successive price changes are independent and identically distributed (Fama, 1991). Consequently, no strategy based solely on historical price patterns can systematically generate excess returns.

Empirical tests of the weak-form EMH often focus on examining the **statistical properties of stock returns**. Fama (1970) advocated the use of autocorrelation analysis, variance ratio tests, and other time series methodologies to detect departures from randomness in returns. Lo and MacKinlay (1988) refined these approaches by identifying subtle serial correlations in U.S. equity returns over short horizons, highlighting that even developed markets may exhibit minor inefficiencies. Campbell, Lo, and MacKinlay (1997) further emphasized that apparent deviations from EMH could result from market microstructure effects, thin trading, or time-varying risk premia, rather than genuine predictability. These insights underscore the importance of robust statistical testing when evaluating weak-form efficiency, particularly in less liquid or frontier markets.

Fama's later work (Fama, 1991) acknowledged that empirical anomalies—such as momentum effects, mean reversion, and post-earnings announcement drift—could appear inconsistent with EMH. However, he argued that these anomalies might reflect compensation for risk, data mining, or model misspecification rather than genuine inefficiency. This perspective is particularly relevant for frontier markets, where structural constraints, limited investor participation, and regulatory gaps can amplify deviations from theoretical efficiency. Bekaert, Harvey, and Lundblad (2007) extend this notion, noting that liquidity constraints in emerging markets can lead to mispricing and serial correlation in returns, even when market participants are rational and well-informed.

In addition to its theoretical contributions, EMH provides a practical benchmark for investment strategy. Under weak-form efficiency, technical analysis—relying solely on past price patterns—is unlikely to yield consistent abnormal returns, whereas fundamental analysis, diversification, and risk-adjusted portfolio optimization become essential tools for investors (Fama & French, 1992). The hypothesis also has important regulatory and policy implications. Efficient markets facilitate capital allocation, reduce transaction costs, and enhance investor confidence by ensuring that securities are fairly priced (Levine & Zervos, 1998). Conversely, market inefficiencies may signal information asymmetry, liquidity constraints, or institutional weaknesses, which policymakers and regulators must address to strengthen market integrity.

In summary, Fama's work provides the **conceptual and methodological foundation** for evaluating weak-form efficiency. By framing the notion that historical prices should not predict future returns in an informationally efficient market, EMH informs both the theoretical justification and the empirical design of the current study. Its emphasis on random walk properties, statistical testing, and the role of market mechanisms provides a framework for assessing whether LUSE adheres to weak-form efficiency, given the unique structural and institutional characteristics of frontier markets in Africa. This foundation justifies the use of Runs Tests, Autocorrelation Tests, and Variance Ratio Tests, ensuring that the analysis is consistent with established financial theory and contemporary empirical practice.

2.2 Market Efficiency in Developed Economies

Empirical studies on market efficiency in developed economies have extensively tested the weak-form Efficient Market Hypothesis (EMH), yielding insights into the predictability of stock returns and the implications for investment strategies.

In developed markets, numerous studies have examined the weak-form EMH, which posits that stock prices reflect all available historical information, rendering technical analysis ineffective. For instance, Borges (2010) conducted an analysis of stock market indexes from the UK, France, Germany, Spain, Greece, and Portugal, covering the period from January 1993 to December 2007. The study employed runs tests and joint variance ratio tests using daily and weekly data. The results indicated that daily and weekly returns were not normally distributed, exhibiting negative skewness and leptokurtosis, and displayed conditional heteroscedasticity. These findings suggest that historical price information alone may not provide a reliable basis for predicting future returns in these European markets.

Similarly, Guidi (2010) tested the weak-form EMH in Central and Eastern European (CEE) equity markets for the period 1999–2009. The study utilized autocorrelation analysis, runs tests, and variance ratio tests. The findings revealed that stock markets in the CEE region did not follow a random walk process, indicating deviations from weak-form efficiency. This suggests that investors in these markets could potentially identify mispriced assets by studying past prices.

The findings from these studies have significant implications for investment strategies in developed markets. The evidence of weak-form efficiency suggests that technical analysis, which relies on historical price and volume data, may not consistently yield superior returns. Investors in such markets may need to consider alternative approaches, such as fundamental analysis or quantitative models, to identify undervalued securities and achieve above-average returns.

The confirmation of weak-form efficiency in developed markets also has policy and regulatory implications. It underscores the importance of transparent and timely dissemination of information to ensure that all market participants have equal access to relevant data. Regulators

in developed economies often implement measures to enhance market transparency and fairness, thereby promoting investor confidence and market stability. While developed markets generally exhibit higher efficiency due to stronger institutional frameworks and deeper liquidity, these findings provide a useful benchmark against which emerging and frontier markets, such as LUSE, can be compared.

2.3 Empirical Studies in Emerging and African Markets

African stock markets present a heterogeneous landscape regarding weak-form efficiency. Mlambo and Biekpe (2007) examined ten African exchanges, finding partial support for weak-form efficiency in Kenya and Zimbabwe, but clear evidence of inefficiency in Botswana and Ghana. Appiah-Kusi and Menyah (2003) further emphasized the importance of institutional quality and market depth, noting that structural and regulatory factors significantly influence price behavior.

Focused studies provide additional context. Chowa, Nyanhete, and Mhlanga (2014) analyzed the Zimbabwe Stock Exchange post-dollarization and found that macroeconomic reforms and institutional changes materially influenced efficiency outcomes. Nyoni and Mlambo (2017) reinforced these findings, showing persistent inefficiencies even after policy improvements. Botswana (Jefferis & Smith, 2019) and Nigeria (Urom & Olowe, 2019) displayed thin trading patterns that caused serial correlations, deviating from the random walk. In South Africa, Odhiambo (2021) demonstrated sector-specific variations in efficiency, highlighting the need to consider market heterogeneity when conducting empirical tests.

Across these studies, **Runs Tests, Autocorrelation Tests, and Variance Ratio Tests** were commonly used to examine price randomness. Daily and weekly data were also emphasized to account for thin trading and nonsynchronous effects. These methodological lessons inform the approach for analyzing LUSE.

2.4 Frontier Market Characteristics and Implications for LUSE

Frontier markets, such as the Lusaka Securities Exchange (LUSE), exhibit distinct characteristics that differentiate them from both developed and emerging markets. These unique features contribute to the operational dynamics of these markets and influence their efficiency.

- **Market Size and Depth:** Frontier markets often have a limited number of listed companies, which can restrict market depth and liquidity. For example, LUSE comprises only 20 listed companies, which can lead to limited investor diversification and increased susceptibility to price manipulation. The concentration of market capitalization in a few firms can amplify the impact of individual stock movements on the overall market index.
- **Trading Volume and Frequency:** The trading volume in frontier markets is typically lower compared to more developed markets. This thin trading can result in higher

volatility and the potential for serial correlations in stock returns. Studies have shown that lower liquidity in frontier markets can lead to less efficient price discovery processes, making it more challenging for investors to assess the true value of securities.

- **Regulatory Environment:** The regulatory frameworks in frontier markets are often evolving and may not be as robust as those in developed markets. Delays in disclosure, weak enforcement of reporting standards, and limited market oversight can contribute to information asymmetry. This environment can hinder the timely dissemination of information, affecting investors' ability to make informed decisions and potentially leading to mispricing of assets.
- **Exposure to External Shocks:** Frontier markets are more vulnerable to external shocks due to their smaller size and limited integration with global markets. Political instability, currency fluctuations, and macroeconomic events can disproportionately impact these markets. For instance, changes in global commodity prices can have a significant effect on the economies of frontier markets that are heavily reliant on exports.

These characteristics suggest that frontier markets like LUSE may be more prone to weak-form inefficiencies compared to larger, more liquid exchanges. The limited number of listed companies, lower trading volumes, evolving regulatory environments, and heightened exposure to external shocks can create conditions where historical price information may not fully reflect all available information, leading to potential predictability in stock returns.

Empirical studies have provided evidence supporting the notion that frontier markets exhibit characteristics that can lead to inefficiencies. For instance, research has indicated that stock returns in frontier markets can be more predictable due to factors such as lower liquidity and less stringent regulatory environments. These findings underscore the importance of conducting rigorous, data-driven assessments of market efficiency in frontier markets to understand the unique dynamics at play and to inform investment strategies.

2.5 Empirical Evidence on LUSE

Existing studies on LUSE are limited and often outdated. Kandiri and Kabiru (2011) and Nwosa (2011) examined weak-form efficiency in selected African emerging markets but provided minimal insight into Zambia. While LUSE trading volumes and investor participation have increased substantially over the past decade, no recent research comprehensively evaluates its weak-form efficiency using both daily and weekly stock price data spanning Jan 2022 – October 2025. The absence of up-to-date empirical evidence constrains investors and policymakers seeking to make informed decisions about market behavior, portfolio strategy, and regulatory reform.

2.6 Methodological Insights

Several methodological lessons emerge from the literature:

- **Multiple Tests Increase Robustness:** Using Runs Tests, Autocorrelation Tests, and Variance Ratio Tests provides a more comprehensive assessment of randomness (Elsayed & Patro, 2008; Lo & Wang, 2000).
- **Logarithmic Returns Stabilize Variance:** Transforming prices to log returns improves accuracy and comparability.
- **Daily and Weekly Data Mitigate Biases:** Combining time horizons addresses thin trading and nonsynchronous trading effects common in frontier markets.
- **Purposive Sampling of Active Stocks:** Selecting actively traded companies ensures data representativeness and reduces bias from illiquid or recently listed stocks.

These methodological considerations guide the current study's approach and ensure alignment with prior best practices.

2.7 Research Gap

Despite the growing interest in the Lusaka Securities Exchange (LUSE) and its increasing trading activity, there remains a significant lack of empirical studies examining its market efficiency, particularly using rigorous econometric and statistical methods. Most existing research on African stock markets has focused on larger or more active exchanges such as those in South Africa, Kenya, and Zimbabwe, with limited attention to smaller frontier markets like Zambia. Furthermore, the studies that do exist on LUSE are either outdated or rely on limited datasets, failing to account for recent structural changes, increased investor participation, and regulatory developments. This gap creates uncertainty for investors, asset managers, and policymakers, as they lack robust evidence on whether stock returns in LUSE follow a random walk consistent with weak-form market efficiency. Addressing this research gap will provide timely empirical insights, helping stakeholders make informed investment decisions and guiding policy measures aimed at enhancing market transparency, liquidity, and resilience.

2.8 Conclusion

The literature demonstrates that while EMH provides a robust theoretical framework, frontier markets such as LUSE present structural, regulatory, and liquidity challenges that may induce weak-form inefficiencies. Empirical evidence from Africa highlights the importance of using rigorous statistical methods and accounting for thin trading, nonsynchronous effects, and market microstructure. Despite recent growth in Zambia's capital market, no current study evaluates LUSE using comprehensive datasets and modern econometric methods. The present study seeks to fill this empirical and methodological gap, providing insights to investors, asset managers, and policymakers regarding the efficiency and investment dynamics of LUSE.

CHAPTER THREE: METHODOLOGY

3.0 Introduction

This chapter outlines the methodological framework adopted to examine the weak-form efficiency of the Lusaka Securities Exchange (LUSE). It presents the research design, models and statistical tests employed, data collection methods, data analysis procedures, and ethical considerations that guided the study. The methodology is designed to ensure the findings are replicable, objective, and consistent with empirical standards used in capital market efficiency research. The approach ensures that the results can be independently verified and compared with similar empirical studies on frontier and emerging markets.

3.1 Research Design

The study adopts a **quantitative research design**, which emphasizes empirical measurement and statistical testing of hypotheses. The objective is to determine whether stock prices on LUSE follow a random walk, as posited by the **Weak-Form Efficient Market Hypothesis (EMH)**.

A descriptive–analytical approach is employed to summarize the behavior of stock returns, identify patterns, and statistically test for dependencies that would indicate market inefficiency. Historical secondary data from listed companies is analyzed through econometric and statistical procedures including the **Runs Test**, **Autocorrelation Test**, and **Variance Ratio Test**.

The quantitative design is appropriate because weak-form efficiency is inherently a measurable concept—it evaluates whether past price information can predict future prices. This approach aligns with prior market efficiency studies such as those by **Worthington and Higgs (2006)**, **Smith (2012)**, and **Lim and Brooks (2011)**, who used time-series econometric methods to test for randomness and predictability in stock returns.

The design also allows replicability, as each step—from data extraction to model testing—is systematically structured and documented.

3.2 Models and Tests for the Study

To assess weak-form efficiency, the study employs three complementary tests frequently used in financial econometrics:

3.2.1 The Runs Test

The **Runs Test** (Bradley, 1968) is a non-parametric method that examines whether the sequence of stock returns is random. It evaluates whether the number of “runs” (sequences of consecutive positive or negative returns) differs significantly from what would be expected in a purely random series. If returns exhibit no predictable patterns, the market can be considered weak-form efficient.

3.2.2 The Autocorrelation Test

The **Autocorrelation Test** measures the correlation between a variable and its lagged values over time. In an efficient market, serial correlations should be close to zero, indicating that past returns do not predict future ones. The **Ljung–Box Q-statistic** is employed to test the joint significance of autocorrelation coefficients across multiple lags (Ljung & Box, 1978). The model can be represented as:

$$r_t = \mu + \phi_1 r_{t-1} + \phi_2 r_{t-2} + \cdots + \phi_p r_{t-p} + \epsilon_t$$

where r_t denotes the logarithmic return at time t , ϕ_i represents the autocorrelation coefficients at lag i , and ϵ_t represents the random error term.

3.2.3 The Variance Ratio Test

The **Variance Ratio (VR) Test** (Lo & MacKinlay, 1988) examines whether variances of multi-period returns are proportional to the variance of one-period returns. In this study, the Variance Ratio Test is implemented manually in R using the method proposed by Lo and MacKinlay (1988), ensuring replicability without reliance on specialized external packages. Under the random walk hypothesis, the variance ratio should equal one. A significant deviation from unity implies mean reversion or momentum, both of which indicate inefficiency. Formally, the test is expressed as:

$$VR(q) = \frac{Var(r_t + r_{t+1} + \cdots + r_{t+q-1})}{q \times Var(r_t)}$$

A variance ratio significantly greater than one implies mean reversion, while a value below one suggests momentum in returns—both indicative of weak-form inefficiency. The test is applied to both individual stock returns and the LUSE All Share Index to detect aggregate efficiency.

3.3 Data Collection

3.3.1 Data Source

The study relies exclusively on **secondary data**, comprising historical daily closing prices of all actively traded companies listed on the LUSE. Data is obtained from the **Lusaka Securities Exchange** ensuring accuracy and completeness.

3.3.2 Data Type and Period

The study utilizes daily closing stock price data from companies listed on the Lusaka Securities Exchange (LUSE) for the period January 2022 to October 2025.

This period captures key phases of Zambia's post-pandemic financial recovery, providing relevant and timely insights into present-day market dynamics. Stock price data is converted into **logarithmic returns** using the formula:

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right)$$

where P_t is the closing price at time t .

3.3.3 Justification of Method

Secondary data is suitable for this study because financial time series are continuously recorded and publicly available, allowing objective, verifiable, and replicable analysis. Furthermore, using official exchange data eliminates response bias and ensures consistency with similar efficiency studies conducted on frontier markets (e.g., **Alagidede & Panagiotidis, 2009; Magnusson & Wydick, 2002**).

3.4 Data Analysis

The analysis is implemented in **R statistical software**, ensuring transparency, reproducibility, and computational accuracy. The process follows a systematic approach:

1. **Data Cleaning and Transformation** – Imported stock prices from Excel, removed missing or zero values, and computed logarithmic returns.
2. **Descriptive Statistics and Visualization** – Calculated mean, standard deviation, minimum, and maximum values, and visualized log returns for all companies.
3. **Normality and Stationarity Testing** – Applied Jarque–Bera, Shapiro–Wilk, and Augmented Dickey–Fuller tests where applicable.
4. **Efficiency Testing** – Conducted Runs Test, Ljung–Box Test, and a manually implemented Variance Ratio Test.
5. **Interpretation and Inference** – Compared empirical findings with the theoretical expectations of the Efficient Market Hypothesis and previous studies on African frontier markets.

Results will be presented using tables, statistical summaries, and visualizations to support clear interpretation. The combined use of non-parametric, parametric, and correlation-based tests strengthens the robustness of conclusions drawn about market efficiency.

3.5 Research Ethics

This research adheres to the ethical guidelines of the **University of Zambia**. All data used is publicly available, thereby eliminating confidentiality concerns. The study ensures integrity through accurate citation of data sources, transparent reporting of procedures, and avoidance of data manipulation. No primary human or institutional participants are involved; hence, issues of informed consent or privacy are inapplicable.

The results will be presented objectively, without bias or selective reporting, and the research will be shared solely for academic and non-commercial purposes.

CHAPTER FOUR: PRESENTATION AND DISCUSSION OF RESULTS

4.0 Introduction

This chapter presents and discusses the empirical findings of the study on the weak-form efficiency of the Lusaka Securities Exchange (LUSE). The analysis applies a range of statistical and econometric tests—namely, descriptive statistics, the Jarque–Bera normality test, the Runs test, the Autocorrelation Function (ACF) test, and the Variance Ratio (VR) test—to evaluate whether stock returns on LUSE follow a random walk. The results are interpreted in relation to the study's objectives and compared with findings from prior studies on African and international markets. Full numerical outputs and diagnostic tables are presented in the appendices.

4.1 Presentation of Results

4.1.1 Descriptive Statistics

Descriptive analysis provides an overview of the statistical properties of daily log returns for the sampled LUSE-listed companies over the study period. Across the sample, mean returns are close to zero, while standard deviations indicate moderate volatility. Most return series exhibit noticeable skewness and excess kurtosis, suggesting deviations from normality. This was further confirmed by the Jarque–Bera test, which strongly rejected the null hypothesis of normal distribution ($p < 0.05$) for all stocks. These findings imply that return distributions on LUSE are leptokurtic and asymmetric, characteristics typical of emerging and frontier markets where thin trading and information asymmetries are common (Magnusson & Wydick, 2002; Mlambo & Biekpe, 2007).

4.1.2 Runs Test for Randomness

The non-parametric Runs test was employed to assess whether stock returns occur in a random sequence. The test results, summarized in **Appendix B**, show that for most of the sampled companies, the calculated z-statistics exceed the critical values at the 5% level, leading to rejection of the null hypothesis of randomness. This indicates that price changes are not entirely independent and that patterns may exist in return movements. The presence of such non-random behavior suggests potential weak-form inefficiency. Similar findings were reported by Appiah-Kusi and Menyah (2003) for selected African exchanges and by Dickinson and Muragu (1994) for the Nairobi Securities Exchange, where serial dependencies in returns were attributed to low market liquidity and delayed information dissemination. These findings indicate that traders on LUSE may occasionally rely on historical price trends or exhibit herd-like trading patterns, reflecting delayed information absorption.

4.1.3 Autocorrelation Function (ACF) Test

Autocorrelation analysis was performed on daily returns up to the tenth lag. The results (see **Appendix C**) show statistically significant autocorrelation coefficients for several short lags, particularly lags 1 and 2, implying short-term dependence among returns. This violates one of the key assumptions of the random walk model, which posits that successive price changes should be uncorrelated. The observed autocorrelation may stem from infrequent trading, non-synchronous price adjustments, or investor herding behavior, as noted in other African studies (Jefferis & Smith, 2005; Magnusson & Wydick, 2002). However, the strength of correlation diminishes with higher lags, suggesting that any predictability is short-lived and quickly corrected by market forces. Overall, the short-term autocorrelation patterns confirm that while the market adjusts to new information, this adjustment is neither instantaneous nor entirely efficient.

4.1.4 Variance Ratio (VR) Test

The Variance Ratio test was conducted as a parametric check for random walk behavior. As shown in **Appendix D**, most variance ratios significantly deviate from unity, particularly for short holding periods ($k = 2$ and $k = 4$). The rejection of the null hypothesis of a random walk in these cases implies that stock returns are not independently and identically distributed. Nonetheless, the degree of deviation from unity lessens as the holding period increases, implying that while short-term inefficiencies exist, they may not persist over longer horizons. Comparable results have been documented by Worthington and Higgs (2006) in their study of 20 emerging markets, where weak-form inefficiency was observed primarily in shorter time intervals.

4.2 Discussion of Results

4.2.1 Objective 1: Assessing Weak-Form Efficiency on LUSE

The collective evidence from all statistical and econometric tests points to a partial rejection of the weak-form Efficient Market Hypothesis (EMH). The combined evidence from the Runs, ACF, and Variance Ratio tests demonstrates deviations from random walk behavior, supporting partial rejection of the null hypothesis. These findings suggest that historical price information retains some predictive power in determining future prices on LUSE.

When compared to international evidence, these results align with those observed in several emerging and frontier markets during the 2010s. For instance, Kiymaz (2015) found similar inefficiencies in selected European emerging exchanges such as Poland and Hungary, where short-term return dependencies were attributed to market thinness and slow information diffusion. Likewise, Appiah-Kusi and Menyah (2003) and Ntim et al. (2007) reported comparable results for African markets, attributing inefficiency to structural and institutional factors, including low turnover, weak regulatory frameworks, and dominance of retail investors.

The evidence therefore suggests that LUSE, like many developing markets, is still evolving toward full efficiency. The short-term predictability detected here does not necessarily imply exploitable

arbitrage opportunities but rather reflects a gradual process of market maturation as infrastructure, liquidity, and information flows improve.

4.2.2 Objective 2: Recommendations for Investors, Asset Managers, and Policymakers

Given the partial inefficiency detected, several implications emerge:

1. For Investors:

Short-term dependencies in stock returns imply that active trading strategies, such as technical analysis and momentum-based approaches, may yield temporary gains. However, these opportunities are likely to diminish as the market becomes more liquid and transparent.

2. For Asset Managers:

The findings underscore the importance of incorporating local market dynamics into portfolio construction. Since LUSE returns exhibit mild predictability, risk-adjusted performance can be enhanced through selective timing and diversification strategies within and across sectors.

3. For Policymakers and Regulators:

The persistence of weak-form inefficiency points to the need for continued reforms aimed at enhancing transparency, liquidity, and investor education. Initiatives such as encouraging institutional investor participation, improving disclosure standards, and leveraging technology to reduce information asymmetries could foster stronger market efficiency over time.

4.3 Summary of Findings

This chapter investigated the weak-form efficiency of the Lusaka Securities Exchange using a combination of descriptive statistics, the Runs test, the Autocorrelation Function (ACF) test, and the Variance Ratio (VR) test. The results from these tests collectively indicate that stock returns on LUSE deviate from the random walk hypothesis, implying partial weak-form inefficiency. Evidence of serial correlation and non-random price changes suggests that past information retains some predictive power for future prices.

These findings are consistent with patterns observed in other African and frontier markets, where limited market depth, low trading volume, and information asymmetry hinder full price adjustment. This confirms that LUSE exhibits characteristics typical of frontier exchanges, including low liquidity and information asymmetry, which contribute to weak-form inefficiency.

However, the inefficiencies observed on LUSE are relatively mild, suggesting a market in transition rather than one that is entirely inefficient. Overall, the results highlight the importance of continued institutional reforms, improved liquidity, and greater investor participation to enhance price discovery and move the market toward higher levels of informational efficiency.

CHAPTER FIVE: SUMMARY, CONCLUSIONS, AND RECOMMENDATIONS

5.0 Introduction

This chapter summarizes the study, draws conclusions based on the empirical results, and provides practical recommendations for various stakeholders. The study examined the weak-form efficiency of the Lusaka Securities Exchange (LUSE) by applying a series of statistical and econometric tests—including the Runs test, Autocorrelation Function (ACF) test, and Variance Ratio (VR) test—to determine whether stock returns follow a random walk.

The objectives of the study were twofold:

1. To assess whether stock returns on the Lusaka Securities Exchange exhibit weak-form efficiency using statistical and econometric tests.
2. To provide recommendations for investors, asset managers, and policymakers based on the findings of the study.

The findings of this research contribute to the growing body of literature on market efficiency in frontier markets and offer important insights into the functioning of Zambia's capital market.

5.1 Summary of Findings

The results indicate that stock returns on LUSE do not fully conform to the random walk hypothesis. Evidence from the Runs test revealed non-random patterns in return sequences, while the ACF and VR tests identified statistically significant short-term autocorrelations and variance deviations from unity. Collectively, these results point toward partial weak-form inefficiency, suggesting that past information may influence future price movements.

The descriptive statistics showed that return distributions on LUSE are non-normal, characterized by skewness and excess kurtosis. These findings align with observations from other African and frontier exchanges such as those in Kenya, Ghana, and Nigeria, where market thinness and information asymmetry have been cited as major constraints to efficiency.

Although inefficiencies were detected, they were moderate in magnitude, implying that LUSE is in a developmental phase rather than being fundamentally inefficient. The gradual improvement in trading systems, disclosure standards, and investor awareness may explain why inefficiencies appear to be declining over time. The gradual improvement observed suggests that reforms in market infrastructure and governance have begun to yield positive effects, even if full informational efficiency remains elusive. This evolution indicates a market moving toward maturity, but still in need of structural and regulatory enhancement to achieve full informational efficiency.

5.2 Conclusions

Based on the results and in line with the study's objectives, the following conclusions are drawn:

Objective 1: To assess whether stock returns on LUSE exhibit weak-form efficiency

The study concludes that the Lusaka Securities Exchange does not fully exhibit weak-form market efficiency. The statistical evidence demonstrates the existence of short-term dependencies and non-randomness in stock returns. This suggests that historical price information retains some predictive value, contrary to the assumptions of the Efficient Market Hypothesis (EMH). Factors contributing to this inefficiency likely include limited market depth, thin trading, delayed information dissemination, and a relatively small investor base dominated by retail participants. Nevertheless, the inefficiency is not pronounced enough to imply market failure; rather, it indicates an evolving market with improving structures.

Objective 2: To provide recommendations for investors, asset managers, and policymakers

From a policy and investment perspective, the findings underscore the importance of enhancing transparency, market liquidity, and investor participation. For investors and asset managers, the partial inefficiency implies the potential—though limited—existence of exploitable short-term opportunities. However, as the market matures, these opportunities are expected to diminish. The overall implication is that LUSE is on a gradual path toward efficiency, contingent upon continuous institutional reforms and modernization.

5.3 Recommendations

Based on the study findings and conclusions, the following recommendations are proposed for various stakeholders:

For the Government of the Republic of Zambia

1. **Enhance Regulatory Frameworks:** Strengthen the legal and institutional mechanisms governing capital markets to ensure timely disclosure, compliance, and enforcement of reporting standards.
2. **Promote Financial Literacy:** Introduce educational initiatives to increase investor awareness and participation, thereby deepening market activity. Incentivize research collaboration between universities and market institutions to generate continuous local evidence on efficiency and market performance.
3. **Encourage Privatization and Listings:** Expand the pool of listed companies through public-private partnerships and targeted incentives, improving market liquidity and diversification.

For the Lusaka Securities Exchange (LUSE)

1. **Improve Market Infrastructure:** Continue modernizing trading systems and enhance transparency through real-time data dissemination.

2. **Increase Trading Frequency:** Implement strategies to reduce market thinness, such as market-making mechanisms that ensure continuous liquidity.
3. **Investor Education Initiatives:** Work closely with professional associations and universities to promote understanding of market dynamics and long-term investing.

For Institutional Investors (NAPSA, PIA, Pension Funds, and Insurance Companies)

1. **Encourage Domestic Investment:** Institutional investors should allocate a higher portion of their portfolios to local equities to increase trading volumes and price efficiency.
2. **Adopt Active Monitoring and Engagement:** As large stakeholders, institutions like NAPSA and PIA can promote better corporate governance and transparency among listed firms.
3. **Leverage Research-Based Strategies:** Given partial inefficiency, institutions should develop data-driven investment models that exploit short-term trends without compromising long-term stability.

For Individual Investors and Asset Managers

1. **Adopt a Mixed Strategy:** Active trading may yield short-term returns in certain securities, but investors should combine it with a long-term portfolio diversification strategy.
2. **Rely on Fundamental Analysis:** Since inefficiency is partial and transient, decisions should be grounded in company fundamentals rather than speculative trading alone.

5.4 Areas for Further Research

This study focused exclusively on testing weak-form market efficiency using secondary data from selected LUSE-listed companies. Future research could explore other aspects of market behavior and efficiency:

1. **Testing Semi-Strong and Strong-Form Efficiency:** Future studies could extend this work by assessing how quickly and accurately new public information is incorporated into prices, providing a fuller picture of market efficiency in Zambia.
2. **Comparative and Sectoral Studies:** Subsequent research could compare efficiency levels across different sectors or between LUSE and other regional exchanges (e.g., Nairobi, Johannesburg, or Dar es Salaam). This would help identify structural and institutional factors that enhance or hinder efficiency in frontier markets.
3. **Behavioral Factors and Investor Psychology:** Exploring how cognitive biases, investor sentiment, and herd behavior affect stock price movements could provide valuable insights into non-fundamental drivers of inefficiency on LUSE.

Ultimately, the Lusaka Securities Exchange reflects a market in transition—imperfect but progressing. With continued institutional reforms, enhanced investor confidence, and technological innovation, it can mature into a fully efficient exchange that supports inclusive and sustainable economic growth in Zambia.

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APPENDICES

The following appendices provide supplementary information supporting the findings and analysis presented in this study.

Appendix I – Data Collection Instruments

This study relied exclusively on **secondary data** sourced from reputable financial databases and institutions. The data consisted of historical stock prices for companies listed on the **Lusaka Securities Exchange (LUSE)**, collected primarily from the **official LUSE website**, and the **Securities and Exchange Commission (SEC) Zambia**.

The dataset included **daily closing stock prices** for all companies listed on LUSE that had sufficient trading history and price continuity over the study period. Companies with irregular trading or suspended listings were excluded to maintain data integrity and analytical consistency.

To ensure data quality, the following preprocessing steps were performed:

1. **Cleaning:** Removal of missing, zero, or extreme outlier values.
2. **Adjustment:** Exclusion of non-trading days and corporate actions (e.g., stock splits or dividends).
3. **Validation:** Cross-checking selected prices with alternative financial reporting sources such as company annual reports and the BoZ time-series database.

The table below summarizes the data sources, periods, and sectors of the analyzed companies.

Table A1.1: Summary of Data Collection Sources

Company Name	Sector	Data Period	Source of Data	Data Type	Remarks
ZANACO Plc	Banking	Jan 2022 – October 2025	LUSE	Daily closing prices	Continuous trading data available
Copperbelt Energy Corporation (CEC)	Energy	Jan 2022 – October 2025	LUSE	Daily closing prices	Highly liquid stock on LUSE
ZAFFICO Plc	Forestry	Jan 2022 – October 2025	LUSE	Daily closing prices	Listed in late 2019; shorter series
Madison Financial Services Plc	Insurance & Finance	Jan 2022 – October 2025	LUSE	Daily closing prices	Limited trading frequency
British American Tobacco (BAT) Zambia	Consumer Goods	Jan 2022 – October 2025	LUSE	Daily closing prices	Thinly traded stock
PUMA Energy Zambia Plc	Energy	Jan 2022 – October 2025	LUSE	Daily closing prices	Moderate liquidity

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National Breweries Plc	Consumer Staples	Jan 2022 – October 2025	LUSE	Daily closing prices	Frequent volatility price
Lafarge Zambia Plc	Construction Materials	Jan 2022 – October 2025	LUSE	Daily closing prices	Active stock pre-merger
Standard Chartered Bank Zambia Plc	Banking	Jan 2022 – October 2025	LUSE	Daily closing prices	Long and stable trading history
Zambeef Products Plc	Agribusiness	Jan 2022 – October 2025	LUSE	Daily closing prices	One of the most actively traded stocks
Zambia Sugar Plc	Manufacturing	Jan 2022 – October 2025	LUSE	Daily closing prices	Consistent data availability
First Quantum Minerals Ltd	Mining	Jan 2022 – October 2025	LUSE	Daily closing prices	Consistent data availability
AECI Mining Explosives Plc	Mining and Manufacturing	Jan 2022 – October 2025	LUSE	Daily closing prices	Relatively illiquid
REIZ Plc	Real Estate	Jan 2022 – October 2025	LUSE	Daily closing prices	Moderate trading activity
Pamodzi Hotels Plc	Hospitality	Jan 2022 – October 2024	LUSE	Daily closing prices	Privatized by ASB Hospitality in 2024
Airtel Zambia	Telecommunication	Jan 2022 – October 2025	LUSE	Daily closing prices	Thinly traded stock
ZCCM Investment Holdings Plc	Mining Investment	Jan 2022 – October 2025	LUSE	Daily closing prices	Major holding company on LUSE
Investrust Bank Plc	Banking	Jan 2022 – April 2024	LUSE	Daily closing prices	Default and delisting in 2023
Shoprite Holdings Zambia Ltd	Retail	Jan 2022 – October 2025	LUSE	Daily closing prices	Consistent trading activity

Note:

All data used in this study were publicly available, secondary in nature, and collected following ethical research standards. The collection procedure ensures that the dataset is reliable, replicable, and appropriate for evaluating weak-form market efficiency on the Lusaka Securities Exchange.

Appendix II – Screenshots for the Codes

This appendix presents selected screenshots of the **statistical and econometric procedures** applied in the analysis of weak-form efficiency on the Lusaka Securities Exchange (LUSE). The screenshots illustrate how the tests were implemented using **RStudio** to ensure transparency and replicability of results.

All codes were written and executed using cleaned log return data for each company. The primary tests conducted include the **Runs Test**, **Autocorrelation Function (ACF) Test**, and **Variance Ratio Test**. Only key excerpts of the codes are displayed here; full tables and detailed numerical results are presented in Appendix III.

A2.1 Runs Test Implementation in R

This test was used to evaluate the randomness of stock return sequences. The R code snippet below shows the function used and the resulting test statistic and p-values for sample companies.

```
65 # =====  
66 #           3. Runs Test  
67 # =====  
68 for (col in numeric_cols) {  
69   cat("\n--- Runs Test for", col, "---\n")  
70   # Make sure we remove NAs before running test  
71   x <- log_returns[[col]]  
72   x <- x[!is.na(x)]  
73   print(runs.test(factor(sign(x))))  
74 }  
75
```

A2.2 Autocorrelation Test Output

ACF plots were generated to visually assess the presence of serial correlation in daily log returns. The plots show autocorrelation coefficients at various lags, with confidence intervals at the 95% level.

```
76 # =====  
77 #           4. Autocorrelation Test  
78 # =====  
79 for (col in names(log_returns)[-1]) {  
80   cat("\n--- Autocorrelation for", col, "---\n")  
81   acf_vals <- acf(log_returns[[col]], plot = FALSE)  
82   print(acf_vals$acf[2]) # lag-1 autocorrelation  
83   cat("Ljung-Box Test:\n")  
84   print(Box.test(log_returns[[col]], lag = 10, type = "Ljung-Box"))  
85 }  
86
```

A2.3 Variance Ratio Test Output

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The Variance Ratio Test (Lo and MacKinlay, 1988) was used to formally test the random walk hypothesis. The screenshot below displays a sample output table from R showing variance ratios, test statistics, and significance levels.

```
87 # =====
88 #           5. Variance Ratio Test
89 # =====
90 variance_ratio_test <- function(x, q = 2) {
91   x <- na.omit(x)
92   n <- length(x)
93   mu <- mean(x)
94   m <- n - q + 1
95   var_1 <- sum((x - mu)^2) / (n - 1)
96   y <- cumsum(x)
97   diff_y <- diff(y, lag = q)
98   var_q <- sum((diff_y - q * mu)^2) / (m - 1)
99   vr <- var_q / (q * var_1)
100   z <- (vr - 1) / sqrt((2 * (2 * q - 1) * (q - 1)) / (3 * q * n))
101   p_value <- 2 * (1 - pnorm(abs(z)))
102   return(data.frame(VR = vr, Z_Stat = z, P_Value = p_value))
103 }
104
```

Note:

Only representative screenshots are included for illustration. Complete output datasets and full code scripts are available upon request.

Appendix III: Detailed Statistical and Econometric Results

This appendix presents the **detailed outputs** from the statistical and econometric tests used to assess weak-form market efficiency on the Lusaka Securities Exchange (LUSE). The results complement the summary findings presented in **Chapter Four**. They include detailed test statistics, p-values, and selected graphical representations of stock return behavior.

Descriptive Statistics of Log Returns

Table A3.1 below presents the detailed descriptive statistics for log returns of all sampled companies. These statistics summarize the mean, standard deviation, minimum, maximum, and kurtosis of daily log returns.

Table A3.1: Descriptive Statistics of Log Returns

Company	Mean	SD	Skewness	Kurtosis
African Explosives and Chemical Industries AECI	0.007107	0.039714	3.933822	21.99401
Airtel Networks Zambia ATEL	0.01036	0.047209	5.850364	41.57714
Bata Shoe Company BATA	0.004647	0.03114	3.494445	31.83892
British American Tobacco Zambia BATZ	0.011966	0.041266	3.550226	15.70994
Copperbelt Energy Corporation CECZ	0.01089	0.040067	3.355849	17.80632
First Quantum Minerals FQMZ	0.092695	0.206401	2.224019	6.578549
Investrust Bank INVE	0.004293	0.027914	6.709766	47.90347
Madison Financial Services MAFS	-0.0016	0.020298	-13.6533	190.2098
National Breweries NATB	-0.00566	0.047684	-9.80732	107.3301
Pamodzi Hotels Plc PMDZ	0.00984	0.127253	13.08462	179.9174
Puma Energy Zambia PUMA	0.006451	0.057325	0.681353	22.79378
Real Estate Investments Zambia REIZUSD	0.003807	0.096655	-6.66561	79.08212
Standard Chartered Bank Zambia SCBL	0.003886	0.064026	0.611187	12.19177
Shoprite Holdings SHOP	0.008581	0.047021	5.855064	46.38376
Zambia Breweries ZABR	0.000111	0.022417	0.039979	32.54153
ZCCM Investments Holdings ZCCM	0.007326	0.036132	2.579754	18.53469

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Zambia Forestry & Forest Industries Corp ZFCO	0.002882	0.020175	7.647227	74.48414
Zambeef Products ZMBF	0.001637	0.04602	1.298736	23.19853
Zambia Metal Fabricators ZMFA	0.013259	0.051228	4.675294	27.39571

Runs Test for Randomness

Table A3.2 shows the results of the **Runs Test** for each stock's log returns, testing the null hypothesis that price changes are random. p-values greater than 0.05 indicate failure to reject the null, supporting weak-form efficiency.

Table A3.2: Runs Test Results (Excerpt)

Company	Z	p_value
AECI	NaN	NaN
ATEL	NaN	NaN
BATA	NaN	NaN
BATZ	NaN	NaN
CECZ	NaN	NaN

Autocorrelation (Ljung–Box) Test

Table A3.3 presents the autocorrelation test results using the **Ljung–Box Q-statistic**, testing whether serial correlation exists in stock returns.

Table A3.3: Ljung–Box Autocorrelation Test Results

Company	Lag1_ACF	p_value
Airtel Networks Zambia ATEL	0.1619	0.1470
Bata Shoe Company BATA	0.1931	0.0411
British American Tobacco Zambia BATZ	0.2266	0.0033
Copperbelt Energy Corporation CECZ	0.1830	0.2319
First Quantum Minerals FQMZ	0.2993	0.8853
Investrust Bank INVE	-0.0241	0.6646

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Madison Financial Services MAFS	-0.0055	1.0000
National Breweries NATB	-0.0093	0.0144
Pamodzi Hotels Plc PMDZ	-0.0066	1.0000
Puma Energy Zambia PUMA	-0.1188	0.2060
Real Estate Investments Zambia REIZUSD	-0.0543	0.9813
Standard Chartered Bank Zambia SCBL	-0.1883	0.0037
Shoprite Holdings SHOP	0.1499	0.0001
Zambia Breweries ZABR	-0.1785	0.0859
ZCCM Investments Holdings ZCCM	0.0224	0.0401
Zambia Forestry & Forest Industries Corp ZFCO	-0.0544	0.9746
Zambeef Products ZMBF	0.2420	0.0341
Zambia Metal Fabricators ZMFA	0.5850	0.0000
Zambia Reinsurance PLC ZMRE	-0.3390	0.0015

Variance Ratio Test Results

Table A3.4 reports the **Variance Ratio (VR) test** outcomes for the same stocks, which directly examine whether returns follow a random walk.

Table A3.4: Variance Ratio Test Results

Company	VR	Z
Airtel Networks Zambia ATEL	0.2817	-6.3927
Bata Shoe Company BATA	0.2459	-6.7114
British American Tobacco Zambia BATZ	0.3533	-5.7554
Copperbelt Energy Corporation CECZ	0.3041	-6.1932
First Quantum Minerals FQMZ	0.2379	-1.8034
Investrust Bank INVE	0.1846	-5.6256
Madison Financial Services MAFS	0.1977	-7.1397
National Breweries NATB	0.1393	-7.6598
Pamodzi Hotels Plc PMDZ	0.1944	-7.1691
Puma Energy Zambia PUMA	0.1383	-7.6683
Real Estate Investments Zambia REIZUSD	0.1825	-7.2752
Standard Chartered Bank Zambia SCBL	0.1911	-7.1991
Shoprite Holdings SHOP	0.2686	-6.5089

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Zambia Breweries ZABR	0.1062	-7.9543
ZCCM Investments Holdings ZCCM	0.2064	-7.0626
Zambia Forestry & Forest Industries Corp ZFCO	0.1751	-7.3408
Zambeef Products ZMBF	0.2928	-6.2939
Zambia Metal Fabricators ZMFA	0.5603	-3.9135
Zambia Reinsurance PLC ZMRE	0.0800	-8.1873

Appendix III (e): Graphical Representations

Below are selected visual outputs generated in **R** to illustrate the behavior of returns and test results.

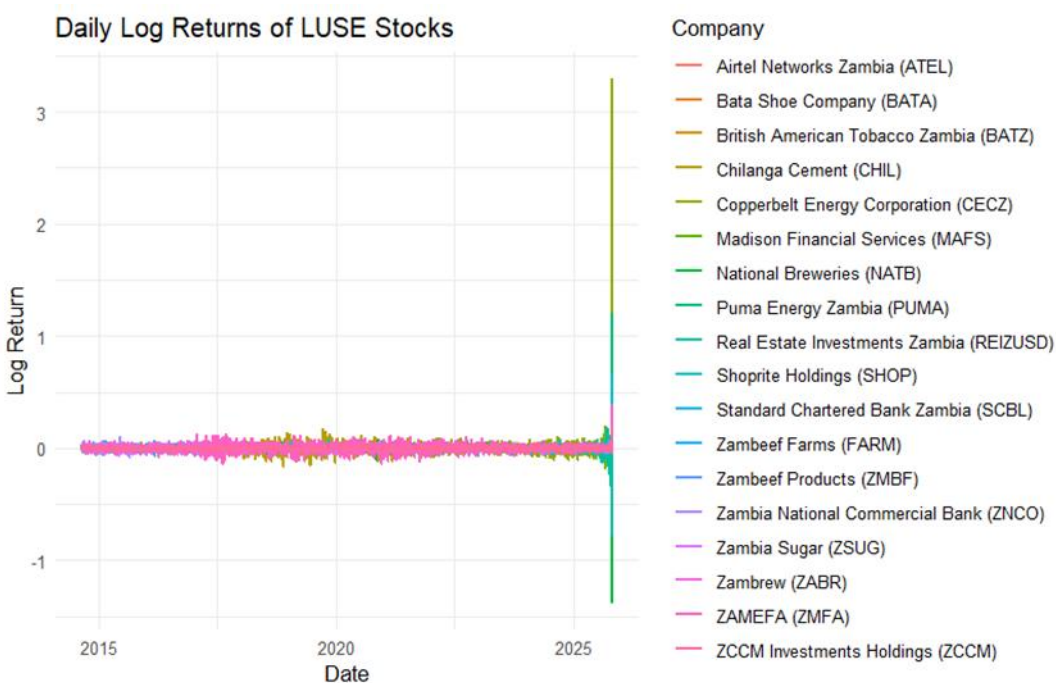


Figure A3.1: Time Plot of Log Returns for Companies

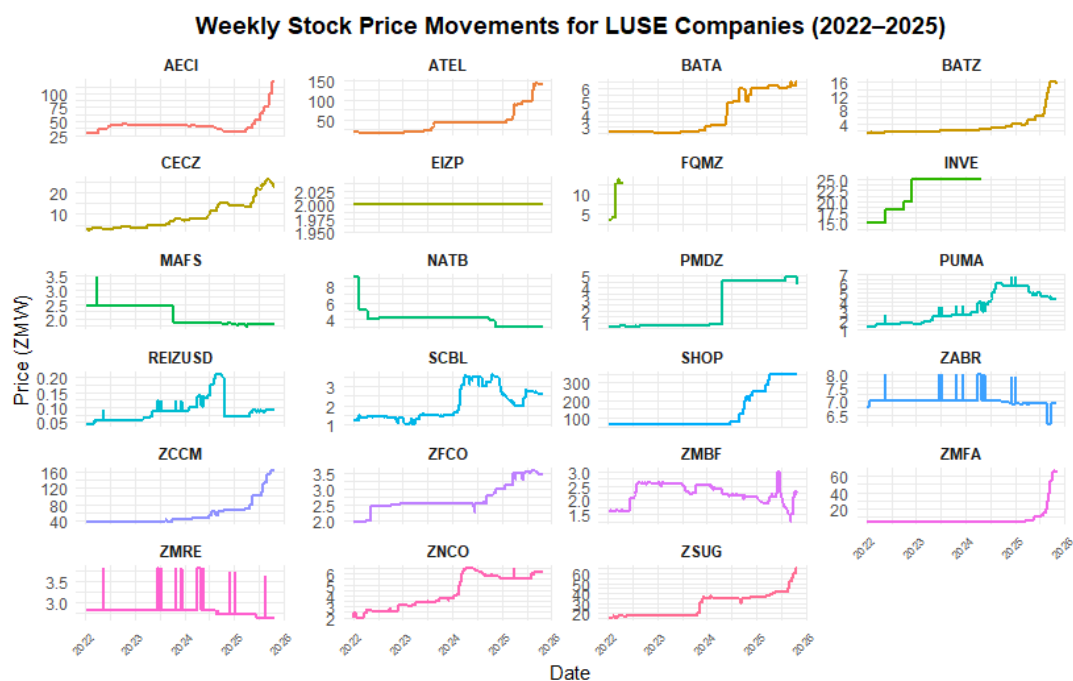
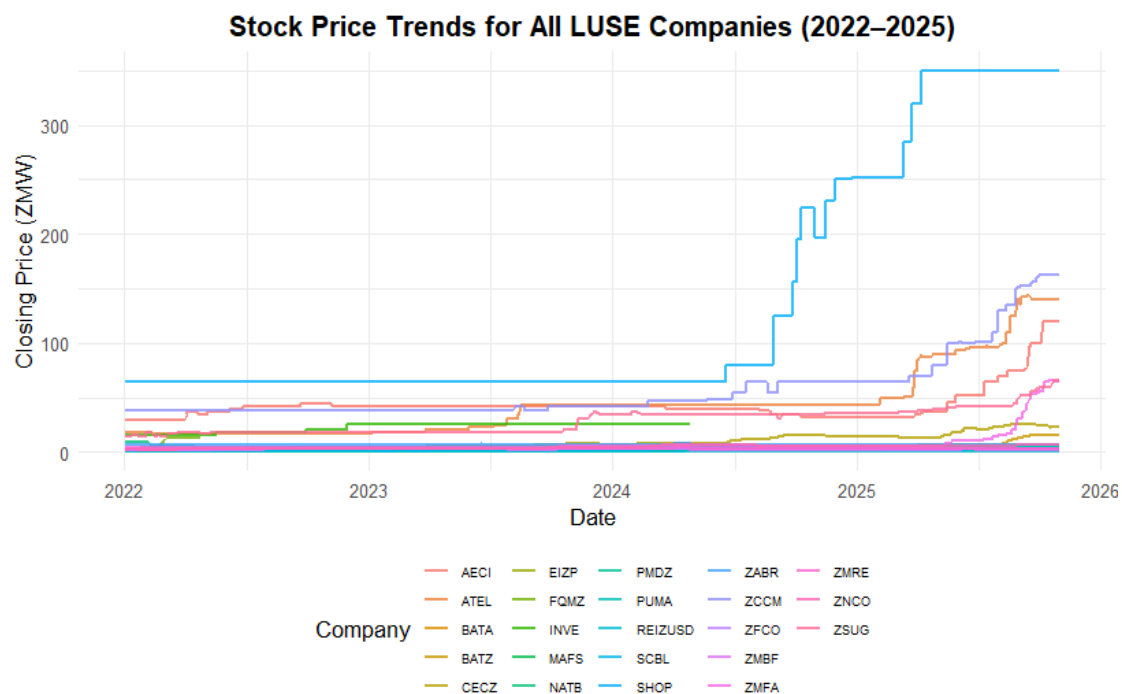
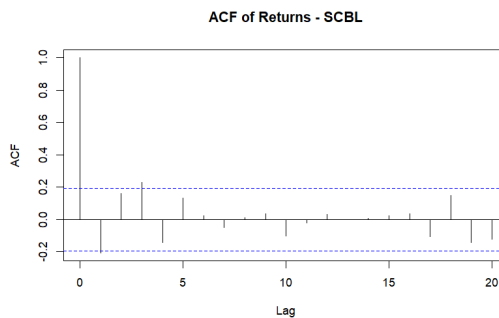
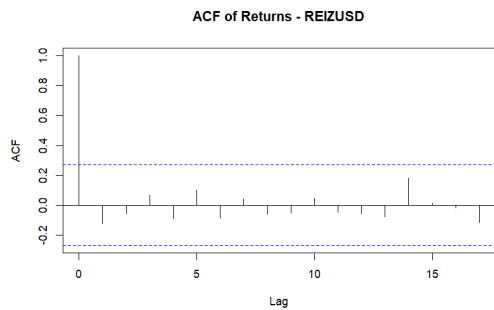
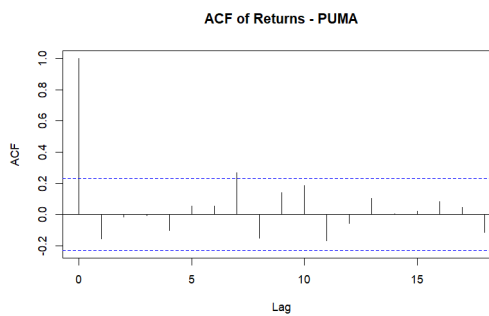
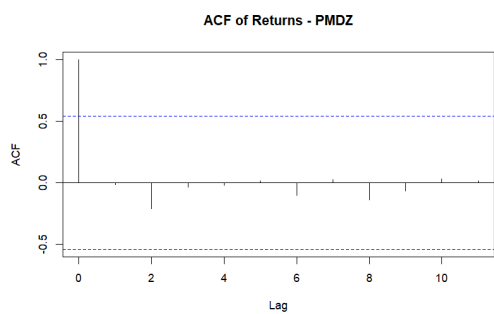
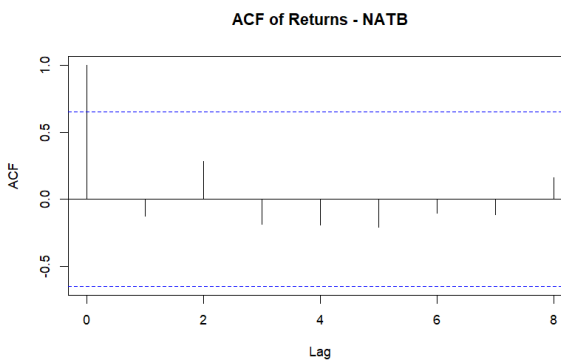
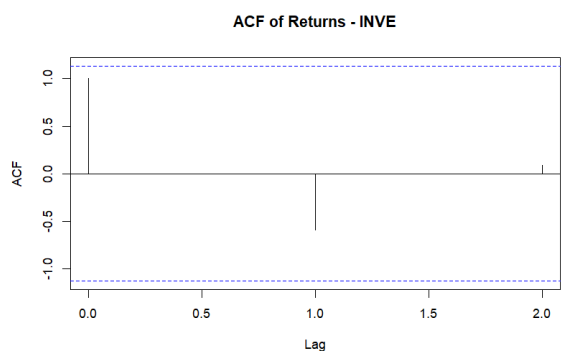
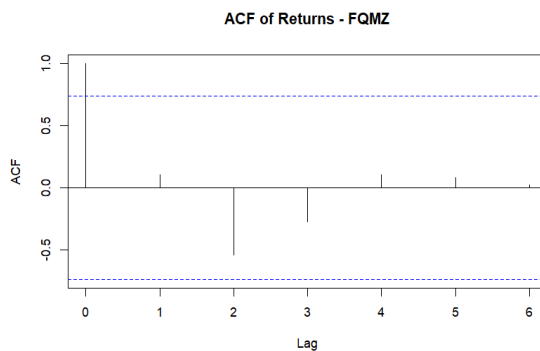
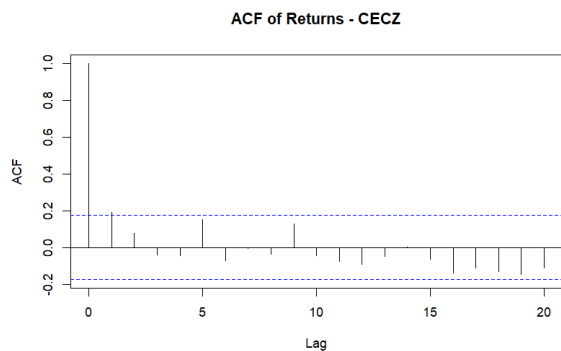


Figure A3.1: Time Plot of Log Returns for Companies

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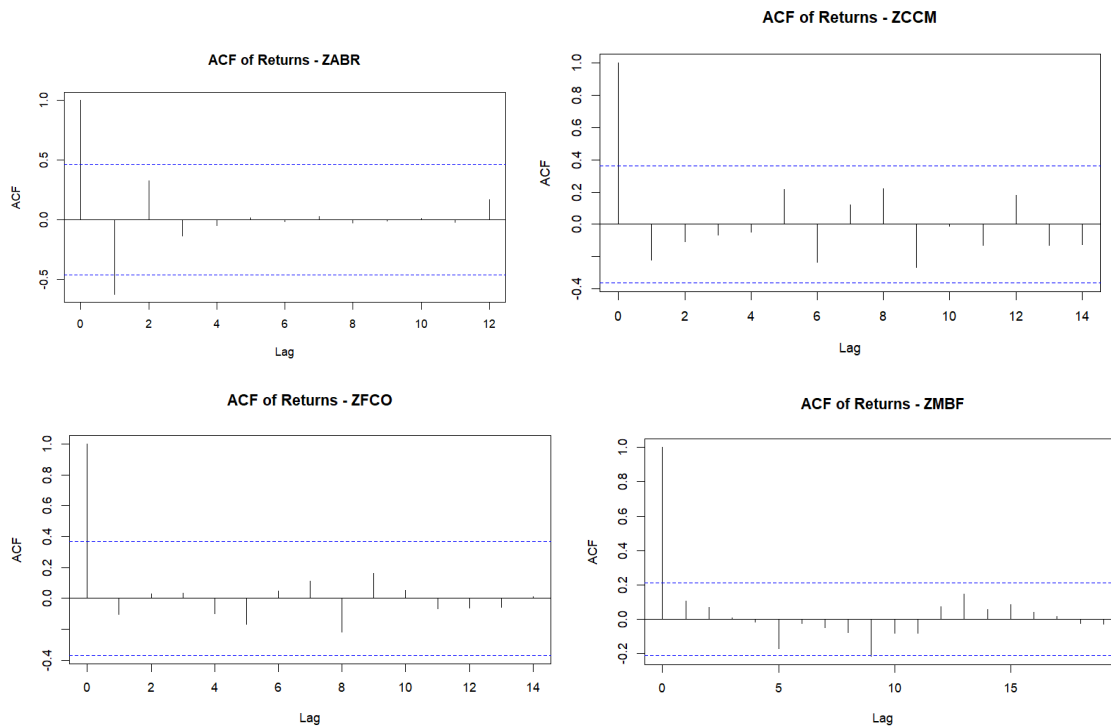


Figure A3.2: Autocorrelation Function (ACF) Plots of Returns

End of Appendices

The materials presented in these appendices provide additional transparency and support for the analyses conducted in this study. They include data sources, coding procedures, and full statistical results that validate the findings discussed in the main chapters. Together, these supplementary materials ensure that the research process is fully replicable and that the conclusions drawn are grounded in rigorous empirical evidence.